Tugas 3

Pengantar Probabilitas (STIK4122)

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Diketahui

Matriks transisi dengan urutan kondisi Tinggi (T), Sedang (S), dan Rendah (R):

$$P = \begin{array}{c} \\ \text{T} \\ \text{S} \\ \text{R} \end{array} \begin{array}{c} \begin{array}{ccc} \text{T} & \text{S} & \text{R} \end{array} \\ \begin{pmatrix} 0{,}4 & 0{,}3 & 0{,}3 \\ 0{,}2 & 0{,}6 & 0{,}2 \\ 0 & 0{,}5 & 0{,}5 \end{pmatrix} \end{array}$$

Distribusi awal:

$$\pi_0 = \begin{pmatrix} 0{,}4 & 0{,}3 & 0{,}3 \end{pmatrix}$$

Catatan: digunakan $\pi_0 = (0{,}4 \;\; 0{,}3 \;\; 0{,}3)$ *sesuai vektor yang tertulis pada soal.*

1. Distribusi kondisi penjualan pada hari berikutnya

Rumus: $\pi_1 = \pi_0 P$

$$\pi_1(T) = (0{,}4)(0{,}4) + (0{,}3)(0{,}2) + (0{,}3)(0) = 0{,}16 + 0{,}06 + 0 = 0{,}22$$
$$\pi_1(S) = (0{,}4)(0{,}3) + (0{,}3)(0{,}6) + (0{,}3)(0{,}5) = 0{,}12 + 0{,}18 + 0{,}15 = 0{,}45$$
$$\pi_1(R) = (0{,}4)(0{,}3) + (0{,}3)(0{,}2) + (0{,}3)(0{,}5) = 0{,}12 + 0{,}06 + 0{,}15 = 0{,}33$$

Jadi,

$$\boxed{\pi_1 = \begin{pmatrix} 0{,}22 & 0{,}45 & 0{,}33 \end{pmatrix}} \qquad (\text{cek: } 0{,}22 + 0{,}45 + 0{,}33 = 1)$$

2. Distribusi kondisi penjualan setelah dua hari

Rumus: $\pi_2 = \pi_1 P = \pi_0 P^2$

$$\pi_2(T) = (0{,}22)(0{,}4) + (0{,}45)(0{,}2) + (0{,}33)(0) = 0{,}088 + 0{,}090 + 0 = 0{,}178$$
$$\pi_2(S) = (0{,}22)(0{,}3) + (0{,}45)(0{,}6) + (0{,}33)(0{,}5) = 0{,}066 + 0{,}270 + 0{,}165 = 0{,}501$$
$$\pi_2(R) = (0{,}22)(0{,}3) + (0{,}45)(0{,}2) + (0{,}33)(0{,}5) = 0{,}066 + 0{,}090 + 0{,}165 = 0{,}321$$

Sebagai pembanding, matriks transisi dua langkah adalah

$$P^2 = \begin{pmatrix} 0{,}22 & 0{,}45 & 0{,}33 \\ 0{,}20 & 0{,}52 & 0{,}28 \\ 0{,}10 & 0{,}55 & 0{,}35 \end{pmatrix},$$

dan $\pi_0 P^2$ memberikan hasil yang sama. Jadi,

$$\boxed{\pi_2 = \begin{pmatrix} 0{,}178 & 0{,}501 & 0{,}321 \end{pmatrix}} \qquad (\text{cek: } 0{,}178 + 0{,}501 + 0{,}321 = 1)$$

3. Peluang berada pada kondisi Sedang setelah dua hari

$$P(X_2 = S) = \pi_2(S) = \boxed{0{,}501}$$

Jadi, peluang perusahaan berada pada kondisi penjualan sedang setelah dua hari adalah 0,501 atau sekitar 50,1%.

4. Distribusi stasioner

Misalkan $\pi = (\pi_T \;\; \pi_S \;\; \pi_R)$ dengan syarat

$$\pi = \pi P \qquad \text{dan} \qquad \pi_T + \pi_S + \pi_R = 1.$$

Dari kolom T:

$$\begin{aligned} \pi_T &= 0{,}4\pi_T + 0{,}2\pi_S + 0 \cdot \pi_R \\ 0{,}6\pi_T &= 0{,}2\pi_S \\ \pi_S &= 3\pi_T \end{aligned} \tag{i}$$

Dari kolom S:

$$\begin{aligned} \pi_S &= 0{,}3\pi_T + 0{,}6\pi_S + 0{,}5\pi_R \\ 0{,}4\pi_S &= 0{,}3\pi_T + 0{,}5\pi_R \\ 0{,}4(3\pi_T) &= 0{,}3\pi_T + 0{,}5\pi_R \\ 0{,}9\pi_T &= 0{,}5\pi_R \\ \pi_R &= 1{,}8\pi_T \end{aligned} \tag{ii}$$

Syarat jumlah sama dengan 1:

$$\begin{aligned} \pi_T + 3\pi_T + 1{,}8\pi_T &= 1 \\ 5{,}8\pi_T &= 1 \\ \pi_T &= \frac{1}{5{,}8} = \frac{5}{29} \approx 0{,}172 \end{aligned}$$

Sehingga

$$\pi_S = 3 \cdot \frac{5}{29} = \frac{15}{29} \approx 0{,}517, \qquad \pi_R = 1{,}8 \cdot \frac{5}{29} = \frac{9}{29} \approx 0{,}310.$$

Cek dengan kolom R:

$$0{,}3\left(\tfrac{5}{29}\right) + 0{,}2\left(\tfrac{15}{29}\right) + 0{,}5\left(\tfrac{9}{29}\right) = \frac{1{,}5 + 3 + 4{,}5}{29} = \frac{9}{29} = \pi_R \quad \checkmark$$

Jadi, distribusi stasionernya adalah

$$\boxed{\pi = \begin{pmatrix} \frac{5}{29} & \frac{15}{29} & \frac{9}{29} \end{pmatrix} \approx \begin{pmatrix} 0{,}172 & 0{,}517 & 0{,}310 \end{pmatrix}}$$